MPI LANCHESTER–OSIPOV

# Modern combat models and scientific novelty

What to add to the Lanchester equations, how to measure intelligence and psychology, how the MPI differs from competitors

Document 8 · Theory and novelty

Measurements: 404 battles of 1900–1973 (CDB90), 10 wars of 2018–2025.

Data cut-off: October 2026 · Open sources
Military power assessment programme for states · project documentation

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# Summary

The Lanchester (1916) and Osipov (1915) equations are a system of differential equations in which a side's losses depend on the enemy's numbers and fire effectiveness. Over a hundred years many extensions have been proposed: randomness of the outcome, space, intelligence, morale, missile salvos, three-sided combat, drones. This document examines which of them are worth adopting and — most importantly — **tests against historical data** which of these ideas are supported.

- **The square law is not supported.** On 404 battles the pre-registered estimate α = 0.65 [0.46; 0.84] excludes both the square (2) and the linear (1) law, and the pre-registered acceptance rule “estimate within [1; 2]” is not met. The exploratory adjustment for combat effectiveness gives 0.97 [0.80; 1.13] — a “nearly linear” result, but it lies outside the protocol and is not used for the conclusion. The popular claim that “a 10% advantage gives 21% more firepower” is not observed in real battles.
- **No form of Lanchester describes the intensity of combat.** Daily losses depend almost not at all on the enemy's numbers (exponent 0.16 instead of 1) and moderately on one's own (0.46); when both sides double, daily losses grow by a factor of 1.54 rather than twofold: only part of the forces takes part in the battle.
- **Intelligence and psychology are measurable.** One step of superiority in leadership, according to the database compilers' assessments, is equal to a numerical advantage of about 9.3 times, in morale — 6.3, in supply — 2.3, in intelligence — 2.2 (2.8 across all years).
- **Combat ends with a collapse of will, not with annihilation.** The median threshold is 16% losses [14%; 18%]; one step of morale raises the threshold by a factor of 1.29. In the combat model the threshold has become a scenario parameter: 16% for a battle, 40% for a campaign.
- **Potential versus effectiveness.** Out of sample, numbers add +0.009 per battle to predicting the outcome of a battle, while the seven effectiveness factors add +0.180 (95% of the total gain). Moreover, across the 10 wars of 2018–2025 the order of the index not only fails to predict whether the advantage will suffice (AUC 0.13) but runs opposite to the outcome: the index measures resources, not effectiveness.
- **Open country-level features capture effectiveness only in part and do not transfer.** GDP per capita, democracy and mechanisation add +0.022 per battle to the prediction of 1900–1973 battle outcomes on top of numbers (8 times less than the compilers’ factors); on the independent UCDP GED base for 1989–2024 the transfer was not confirmed (Spearman +0.248, n = 24).
- **The scientific novelty of the MPI lies in testability:** translating the index into force with out-of-sample validation according to pre-registered protocols, including negative results. The drone extension of Lanchester was published (Gogerty, 2026) without validation against data; we tested its key claim on Oryx losses out of sample, and it **was not confirmed** (section 2.3).

# 1. The classical model and its validation

## 1.1. Lanchester and Osipov equations

Let A and B be the numbers of the sides, a and b their fire effectiveness (targets killed per unit time per unit of force). The square law (aimed fire, each sees a target):

```
dA/dt = −b·B,   dB/dt = −a·A   ⇒   a·A² − b·B² = const
```

A side's strength is proportional to effectiveness and to the **square** of its numbers: a 10% superiority in numbers gives a 21% advantage. The linear law (area fire, no visible target) — losses are proportional to the density of one's own forces:

```
dA/dt = −b·A·B,   dB/dt = −a·A·B   ⇒   strength ∝ effectiveness × numbers
```

The MPI uses the Helmbold generalisation with exponent α (α = 2 — the square law, α = 1 — the linear law in terms of the loss ratio):

```
dA/dt = −a · A^(1−α/2) · B^(α/2),   dB/dt = −b · B^(1−α/2) · A^(α/2)
ln(L_off / L_def) = ln d + ln(e_def / e_off) + (α − 1) · ln(N_def / N_off)
```

## 1.2. What the data showed

| Source | Data | Conclusion |
|---|---|---|
| Rubio-Campillo (PLoS ONE, 2016) | 1,080 battles of 1620–1905, Bayesian model selection (ABC) | neither the linear nor the square law fits; the “fatigue” model fits best |
| MPI, CDB90 (protocol 567224b) | 404 battles 1900–1973 | α = 0.65 [0.46; 0.84]; $R^2$ = 0.04; attackers win in 68%; the loser loses a median of 10% |
| MPI, CDB90 adjusted for effectiveness | 377 battles | α = 0.97 [0.80; 1.13]; d ≈ 1.0–1.2; one step of effectiveness: attacker's losses × 0.42, odds of victory × 3.0 |
| MPI, 10 wars 2018–2025 (protocol 5ed015a) | index based on data for the year before the war | all variants are worse than the uniform forecast (-14.6): -64.4 at α 1.0; model error σ = 2.0 |

> **Conclusion.** The apparent historical “α between 1 and 2” is mostly explained by selection: attackers are qualitatively stronger. The exploratory effectiveness-adjusted estimate (0.97) gives nearly a linear law: numerical advantage counts proportionally, not squared, while quality (effectiveness, leadership, morale) weighs more than numbers. But this is an exploratory result, not a pre-registered one: the protocol accepted only “both laws rejected”.

# 2. Modern extensions of the model

Below are the main development directions with an assessment of whether they should be included in the MPI. Criterion: an extension is accepted if its parameters can be measured and tested against data.

| Direction | Essence | In the MPI | Decision |
|---|---|---|---|
| Helmbold, variable coefficients | effectiveness depends on the force ratio | yes (combat model) | — |
| Stochastic models (Markov chains) | probability of victory instead of the mean outcome | Monte Carlo of parameters and model error σ | sufficient |
| Breaking point, tolerance to losses (Atkinson, Kress, MacKay 2021; Shumov 2016) | combat ends with a collapse of will | threshold bp | measured (section 4.3) |
| Mixed law, intelligence (Deitchman 1962, Shaffer 1967, Coulson 2019) | aimed fire versus area fire | none | validated (section 4.1) |
| Hughes salvo model | missile combat: salvo versus interceptors | none | borrow for the navy and air defence |
| Three-sided combat (Kress et al., 2018) | third force | none | through drones (section 6) |
| Spatial models (PDE), swarms | movement and density across territory | none | later: no data for validation |
| Fractional order (“memory”) | fractional derivatives | none | no: no empirical basis |
| SINDy — deriving equations from data (Chung, Jeong 2024) | the system is fitted to Oryx loss series | none | no out-of-sample validation — a gap |
| Dupuy QJM/TNDM | combat effectiveness multiplier CEV | Q and M roles | multiplier E (section 4.2) |
| Economics of attrition | output versus expenditure | none | borrow (section 6) |
| Gogerty 2026 «Square Law → Swarm Law» | drones — a third force, cost exchange, EW, depth of production | — | idea published, no validation |

## 2.1. Salvo model (Hughes)

For missile combat, where fire is discrete and interception matters no less than the strike, Hughes proposed the salvo model. B's losses per salvo:

```
ΔB = (α·A − y·B) / w   (not less than 0)
α — missiles in a salvo per ship A, y — interceptors per ship B, w — hits to put a ship
out of action
```

The model is suitable for navies and for the standoff “strike drones and missiles against air defence”: the result is determined by the depth of the interceptors' ammunition load and the cost of a single interception, not by the number of platforms.

**Layered air defence and target speed (implemented in the prototype calculator, `core/lanmil/salvo.py`).** Targets pass through the layers in order, and each layer spends stock only on the targets that reached it; the effectiveness of an interceptor falls with target speed above its reference speed:

```
p_eff = p0 for v ≤ v_ref,  otherwise  p0 · (v_ref / v)^g
a target passes a layer with probability (1 − p_eff)^n,  n — whole interceptors per
target
stock_j(t+1) = stock_j(t) − k_j · (number of targets reaching layer j) + output_j
```

The exponent g = 0.5 was chosen roughly from two claimed class D points (piston-engine Shaheds are intercepted in 92–98% of cases, jet-type ones in about 60%; the Ukrainian Air Force, FT, September 2026) and is **not a calibration**: g and the reference speeds are set by the user. The default scenario is

SAM missiles as the outer layer and interceptor drones as the inner one (prices from open estimates: an interceptor of about $1 million and about $5 thousand against a target of $30 thousand). The result is a demonstration: against a fast target (500 km/h instead of 180) the effectiveness of the inner layer falls from 0.70 to 0.49 and day-one leakage rises from 2.9 to 5.7 out of 100; the stock of the expensive outer layer runs out by the end of the month in both variants, while the cheap inner one accumulates, and the cost exchange stays at about 5 : 1 in favour of the offence (the defence spends five times more). The proof that a uniform distribution of interceptors over targets is optimal (arXiv 2608.31143, a preprint) means that the calculation is the best case for the defence.

## 2.2. Three-sided combat and drones

Kress, Caulkins, Feichtinger, Grass and Seidl (EJOR, 2018) solved the Lanchester system for three sides. Gogerty (SSRN 6881358, 2026) uses it to introduce mass-produced UAVs as a third “force” with effectiveness depending on cost exchange and EW. His main conclusion: in a war with mass drones, the outcome is determined by the depth of production. This is a scenario analysis without validation against data.

## 2.3. Validation against the Russia–Ukraine war: does drone output determine the outcome

A testable claim: in a war with mass drones, the loss ratio is determined by the relative output of cheap means of destruction. Operationally: ln r = c + b·x, where r is the ratio of Russia's and Ukraine's losses over a year (all equipment, Oryx), x = ln(Ukraine's output ÷ Russia's output) according to the parties' statements. The protocol was fixed before computation (commit 061ed46): b is estimated from the 2023 → 2024 transition, the forecast is made for 2025, which does not participate in the estimation; the benchmarks are “nothing changed” and “constant d” over 2022–2024.

| Year | Russian losses (units) | Ukrainian losses (units) | Ratio r | x = ln(P*Ukr* ÷ *P*Rus) |
|---|---|---|---|---|
| 2022 (from 24.02) | 8560 | 2677 | 3.20 | — |
| 2023 | 5168 | 2237 | 2.31 | 1.74 |
| 2024 | 5984 | 2486 | 2.41 | 0.38 |
| 2025 | 4100 | 3537 | 1.16 | 0.69 |

**Result: the hypothesis is not supported.** Between 2023 and 2024 Ukraine's relative output fell (x from 1.74 to 0.38), while the loss ratio did not change (2.31 → 2.41), so b = -0.03. The forecast for 2025 is r = 2.38, actually 1.16; the direction of change is predicted incorrectly. Mean error |Δ ln r|: 0.72 for the hypothesis, 0.73 for “nothing changed”, 0.81 for constant d — the hypothesis is no better than the null benchmarks. With alternative Ukrainian output of 3 million in 2025, it is also not supported (error 0.73).

> **What this means.** in 2025 the loss ratio fell sharply — from 2.4 to 1.2, by autumn 2025 to about 0.9 (the parties' losses equalised), — whereas the stated relative output changed little. The causes have not been established. Caveats: this is a single test point; statements about output are incomparable by definition and political; Oryx publications at the end of 2025 may have been delayed; the ratio depends on who is attacking. The result concerns the simple relationship “stated output → loss ratio”, not the role of drones as a whole.

## 2.4. Why the hypothesis did not work: yield per drone and the composition of losses (branch B)

The next step is to understand the cause rather than look for another fit. Status of the analysis: **exploratory, outside the protocol** (the loss ratio for 2025 was already known from the test in 2.3). Russia's losses are compared with the claimed drone output of Ukraine in two independent sources — Oryx and WarSpotting (records by date of loss).

| Year | Ukrainian drone output, million (claimed) | Russian losses, Oryx | Russian losses, WarSpotting | Oryx per million drones | WarSpotting per million drones | Share of heavy equipment | Share of light vehicles |
|---|---|---|---|---|---|---|---|
| 2023 | 0.80 | 5168 | 4315 | 6460 | 5394 | 64% | 19% |
| 2024 | 2.20 | 5984 | 6296 | 2720 | 2862 | 72% | 17% |
| 2025 | 4.00 | 4100 | 3464 | 1025 | 866 | 58% | 29% |

The yield of losses per unit of claimed output fell by a factor of 6.3 according to Oryx (6460 → 1025 per million) and by a factor of 6.2 according to WarSpotting. The main point: between 2024 and 2025 the claimed Ukrainian output rose from 2.20 to 4.00 million, while the number of recorded Russian losses **fell** — by 31% according to Oryx and by 45% according to WarSpotting. Any model of the form “losses are a non-decreasing function of output” with constant parameters (including the saturation model L = T·(1 − e^(−y·D/T))) is rejected by this in both independent sources.

In the WarSpotting records the share of heavy equipment (tanks, infantry fighting vehicles, self-propelled artillery) fell from 72% in 2024 to 24% in 2026 (as of 20.10.2026), while the share of light vehicles (transport, light mobility vehicles) rose from 17% to 48%. This is consistent with the descriptions by RUSI and Defence Studies: drones account for about two thirds of losses, but only 20–40% of them reach the target, and the adversary adapts — it disperses and replaces armour with light vehicles and infantry assaults.

> **Conclusion.** Claimed output is a poor input parameter: losses are limited not by the number of “shooters” but by the number of vulnerable targets and by the adversary's adaptation. A proper attrition model is of the predator–prey type: losses = output × share arriving × share of hits × **the share of targets still present at the front**; the last factor itself declines. This is a testable hypothesis for a future period (the share of heavy equipment in the loss records will not return to the 2024 level), but not an accepted model: there are no independent monthly series of the number of targets at the front in open data. Caveats: output is claims by the parties, not comparable in definition; Oryx and WarSpotting record only what is confirmed by photo and video; 2026 is incomplete.

# 3. How models account for intelligence and psychology

## 3.1. Intelligence — transition between laws

Let i be the share of fire at reconnoitred targets (0…1). Then losses are a mixture of aimed and area fire:

```
dA/dt = −b·B·[ iB + (1 − iB)·A/A₀ ]
dB/dt = −a·A·[ iA + (1 − iA)·B/B₀ ]
i = 1 — the square law, i = 0 — the linear law (area fire)
```

In this entry, intelligence is not merely a force multiplier but what determines which law governs the battle. Hence Coulson's (2019) well-known result: under the square law, superiority in intelligence compensates for numbers only through the square root. But this is a consequence of the square law itself; under the linear law (as in our data), intelligence compensates for numbers one to one.

## 3.2. Psychology — will as a state variable

```
dm/dt = −μ·(relative loss rate) − ν·(strikes on the rear) + ρ·(1 − m)
fire is delivered by A·m combatants (Shumov: not incapacitated and not refusing to
fight)
a side ceases combat when m < m*  (will threshold, tolerance to losses)
```

Atkinson, Kress and MacKay (Operations Research, 2021) showed on such a model that more units **and** better weapons (a higher kill probability) are more advantageous than improved targeting and the removal of deployment constraints. The widespread retelling that “numbers matter more than new models” does not correspond to this work.

# 4. Measurements on CDB90 battles

The CDB90 database (Concepts Analysis Agency, 1990) contains numbers, losses, duration and the HERO compilers' assessments — in whose favour were intelligence, morale, leadership, training, equipment, supply and surprise (scale −2…+2). The protocol was fixed by commit 65e1a59 before computation; the main sample is 404 battles of 1900–1973, the checks are all years (643) and battles with reliable numbers (485). The intervals are 95%, bootstrapped over battles.

> **Data trap.** In the widespread “tidy” version of CDB90 all negative codes (assessments in favour of the defence, attacker defeats) are recorded as blanks. The MPI reads them from the source tables; without this the sample loses up to 20% of battles and becomes biased.

## 4.1. The fire law: what daily losses depend on

```
ln(L/T) = c + p · ln N_enemy + q · ln N_own + defence type + role
```

| Sample | p (opponent) | q (own) | p + q | Expected |
|---|---|---|---|---|
| 1900–1973 | 0.16 [0.06; 0.27] | 0.46 [0.36; 0.56] | 0.62 [0.55; 0.70] | square: 1 and 0; area: 1 and 1 |
| with front width | 0.22 [0.10; 0.35] | 0.49 [0.37; 0.61] | 0.71 | density adjustment |
| all years | 0.15 [0.05; 0.23] | 0.46 [0.37; 0.56] | 0.61 | |
| reliable | 0.18 [0.07; 0.29] | 0.44 [0.33; 0.54] | 0.62 | |

Neither the square law nor the area-fire law describes how much a side loses per day: the enemy's numbers have little effect (p = 0.16), one's own have a moderate effect. The sum p + q = 0.62 < 1 means that only part of the forces takes part in the battle — this is the deployment constraint from the work of Atkinson, Kress and MacKay. For the outcome, the loss ratio matters more, and it is consistent with the model: α = 1 + p − q ≈ 0.71. The share of aimed fire i from section 3.1 cannot be determined this way (it requires p ≈ 1) — this is an honest negative result.

## 4.2. Intelligence and psychology in numerical advantage

The loss exchange and the probability of victory were estimated simultaneously across all seven factors (377 battles), accounting for the force ratio and the type of defence. The equivalent is how many times one must outnumber the enemy to obtain the same increase in the odds as one step of a factor gives.

| Factor (step) | Odds of victory × | Equal to the advantage, times | All years, times | Attacker losses × |
|---|---|---|---|---|
| Leadership | 13.4 [7.1; 50.9] | 9.3 [4.2; 67.7] | 14.7 | 0.61 |
| Morale | 8.5 [4.4; 32.8] | 6.3 [2.8; 47.0] | 10.7 | 0.77 |
| Supply | 2.6 [1.1; 11.3] | 2.3 [1.1; 13.8] | 3.0 | 1.60 |
| Intelligence (significant in all years) | 2.5 [0.9; 18.1] | 2.2 [0.9; 18.0] | 2.8 | 0.81 |
| Surprise | 1.8 [1.1; 3.1] | 1.6 [1.1; 3.1] | 1.4 | 0.90 |
| Equipment (not significant) | 0.8 [0.1; unbounded] | 0.8 [0.1; unbounded] | 1.4 | 0.71 |
| Training (not significant) | 0.7 [0.4; 1.2] | 0.7 [0.4; 1.2] | 1.4 | 0.65 |

The working form for the combat model is the side's effectiveness multiplier:

```
E = Π kf ^ (step f),   k: leadership ≈ 9, morale ≈ 6, supply ≈ 2.3, intelligence ≈ 2,
surprise ≈ 1.6
```

> **Caveat.** The factor assessments were assigned by the HERO compilers after the battles, when the outcome was known. The relationship “won, therefore leadership was better” inevitably overstates the effects, so the multipliers are an upper bound. Training and equipment are not separately distinguishable when the other factors are controlled for.

## 4.3. Will threshold

The losing side collapsed at its own loss share, the winning side did not (its threshold is above the observed share). The threshold is modelled by a lognormal distribution, estimated by maximum likelihood with censoring.

| Model | Median threshold | Spread s | Effects |
|---|---|---|---|
| C0 — common threshold | 16% [14%; 18%] | 1.26 | — |
| C1 — attacker / defender | 16% / 16% | 1.25 | defence 0.05 [-0.18; 0.26] — no difference |
| C3 — + morale (outside the protocol) | 15% [12%; 18%] | 1.26 | one step of morale — threshold × 1.29 [1.11; 1.47] |
| all years, C0 | 21% [19%; 23%] | 1.17 | |

The protocol's model C2 contained the covariate “loss rate” ln(f/T), which includes the loss share f itself — its coefficient is mechanically positive and cannot be interpreted; C3 is the same model without it (an analysis outside the protocol).

```
m* ≈ 0.16 · 1.29^(morale)
```

The candidate bp = 0.16 was tested on 10 wars of 2018–2025: logarithmic score -66.5 versus -66.0 with the previous 0.4 — no worse, but no better either. **Decision (13.10.2026): the threshold is a scenario parameter.** A separate battle (days) — 16%, a campaign (months) — 40%, by default; in the prototype — the “Break threshold” switch.

## 4.4. Potential × effectiveness: what explains the outcome (branch A)

Hypothesis: the strength of a side is the product of its potential (numbers) and its effectiveness. The test is cross-validation (5 folds × 40 repeats) on 343 battles with no gaps in the factors; the measure is the out-of-sample logarithmic score per battle (higher is better); the rule of protocol ddf406d was fixed before the calculation: the gain from the factors over numbers must be at least 0.10 and the confidence interval must lie above zero.

| Model | Features | Out-of-sample score | Gain over the previous model |
|---|---|---|---|
| A0 | type of defence | −0.641 | — |
| A1 | + numbers (ln N attacker / N defender) | −0.632 | +0.009 [+0.003; +0.012] |
| A2 | + seven factors (intelligence, morale, leadership, training, technology, supply, surprise) | −0.452 | +0.180 [+0.149; +0.193] |
| A3 | factors without numbers | −0.474 | +0.167 over A0 [+0.144; +0.180] |

Numbers add almost nothing to predicting the outcome (+0.009 per battle); the effectiveness factors add roughly 20 times more (+0.180); the factors account for 95% of the entire gain over the type of defence. For the loss exchange ln(Latt / Ldef) the share of variance explained out of sample is 0.00 for numbers and 0.23 with the factors. The protocol rule is met, with a caveat: the factor codes were assigned by the compilers after the battles, so the contribution of the factors is an upper bound; the contribution of numbers cannot be overstated.

```
F = M · E,   M — potential (numbers × equipment quality),   E = Π kf ^ (step of f) — effectiveness (section 4.2)
```

Implication for the MPI: the index mainly measures M. To predict outcomes it needs the multiplier E, and the sources for E in modern wars are exactly what open data do not provide (training, leadership, morale). This justifies the expert panel (document 7, modules M5–M6) as a necessary part of the method rather than an ornament.

## 4.5. Order of the index and the balance of forces (branch G)

If κ is not identifiable (section 1.2), can the order of the index at least be trusted? The area under the ROC curve (AUC) does not depend on κ: the test is whether ln(ratio) is higher in the cases where the advantage sufficed than in those where it did not. Ten wars of 2018–2025, candidate θC.

| Case | Advantage by the index | Required | Did the advantage suffice? |
|---|---|---|---|
| Israel and the United States against Iran, June 2025 | 0.53 : 1 | 1.13 : 1 | yes, it did |
| Rwanda (via M23) against DR Congo, 2025 | 1.07 : 1 | 1.00 : 1 | yes, it did |
| Israel against Iran in Syria, 2018 | 1.14 : 1 | 1.30 : 1 | yes, it did |
| Rwanda (via M23) against DR Congo, 2021–2022 | 1.19 : 1 | 1.00 : 1 | no, it did not |
| China versus India, Ladakh, 2020 | 1.49 : 1 | 1.00 : 1 | no, it did not |
| Russia versus Ukraine, 2022 invasion | 1.77 : 1 | 1.49 : 1 | no, it did not |
| Azerbaijan against Armenia, Karabakh, 2020 | 1.84 : 1 | 0.87 : 1 | yes, it did |
| India against Pakistan, May 2025 | 3.94 : 1 | 1.30 : 1 | no, it did not |
| Pakistan versus Afghanistan, 2024 | 6.84 : 1 | 1.30 : 1 | no, it did not |
| Russia versus Ukraine, 2023 offensive | 8.83 : 1 | 1.15 : 1 | no, it did not |

AUC = 0.13 (the protocol expected at least 0.70), permutation probability in favour of superiority 0.98: the rule “the order of the index is informative” is not met. Yet the value is not “about 0.5” but substantially lower — the order of the index is **opposite** to the outcome. Outside the protocol (after the calculation, hence exploratory): the exact permutation p for the inverse relationship is 0.033 (one-sided) and 0.067 (two-sided). The five attackers with the largest advantage by the index (India 2025, Pakistan 2024, Russia 2023 and 2022, China against India 2020) did not achieve their aims; Israel, whose index is weaker or equal, did.

> **Caveat.** Ten cases are few and not independent (Russia 2022 and 2023 are one war; the two Rwanda cases are one theme); the inverse order is not a hypothesis of the protocol but an observation after the calculation. The conclusion is cautious: the order of an index of resources does not predict the success of an offensive, because the index measures size, not effectiveness (section 4.4). For forecasting wars the index should be presented as an ordinal estimate of resources, without conversion into “strength” and without the promise that a larger advantage means a better chance.

## 4.6. Effectiveness from open country-level features (CDB90 and UCDP GED)

Section 4.4 showed that effectiveness factors explain the outcome, but their codes were assigned by the compilers after the battles. Can they be replaced by features known before the battle? Protocol acd9792 was fixed before the calculation. Side features (attacker minus defender, last value within 5 years before the year of the battle): ln GDP per capita (Maddison), electoral democracy (V-Dem), ln energy consumption per soldier (CoW NMC — a measure of mechanisation), combat experience in the same war. Sample: n = 354 battles of 1900–1973 where both sides are matched to countries; the measure and cross-validation are as in Section 4.4.

| Model | Features | Out-of-sample score | Gain over the previous model |
|---|---|---|---|
| E0 | type of defence | −0.687 | — |
| E1 | + numbers | −0.679 | +0.008 [+0.004; +0.012] |
| E2 | + GDP per capita, democracy, mechanisation | −0.657 | +0.022 [+0.015; +0.027] |
| E3 | + combat experience | −0.656 | +0.001 [−0.004; +0.004] |

Rule 1 (country features add at least 0.02 on top of numbers) is **met**: +0.022. This is the project's first positive predictive finding on top of numbers, but it is roughly 8 times smaller than the upper bound from the compilers' factors (+0.180). Rule 2 (combat experience beyond the country) is not met. The out-of-sample share of variance explained in the loss ratio rose from 0.01 (numbers) to 0.20 (country features) — the country explains a fifth of the spread, as in the literature (Biddle, Reiter and Stam): losses are determined by how forces fight, not by how many there are.

**Independent check on UCDP GED v25.1.** The loss-exchange model (three features, no intercept, trained on CDB90 only, n = 354) was applied to interstate dyad-years of 1989–2024 (losses on both sides, at least 25 deaths): 24 dyad-years in 9 dyads, 12 of them India — Pakistan. The protocol with its rule (rank correlation of at least 0.30 and the lower bound of a dyad bootstrap above zero) was committed before the calculation. Result: Spearman +0.248 [−0.596; +0.627], the sign of the exchange guessed in 50% of cases; on the modern wars of 2018–2025 the models trained on CDB90 gave AUC 0.50 (E1) and 0.38 (E2) against "superiority sufficed". **The transfer is not confirmed.**

**A second independent base — Correlates of War wars (Inter-State War Data v4.0).** ACLED is available only with a key, so the COW interstate wars with participants' losses were used: the unit is a war, features are taken for the lead states of the sides, the model is the same (trained on CDB90), and the protocol was fixed before the calculation. The main sample is wars since 1974, after CDB90 ends (17 wars): Spearman +0.520 [−0.014; +0.837], the sign guessed in 59% of cases. The direction matches CDB90 (the richer and more democratic side bears relatively smaller losses), the lower bound of the interval is slightly below zero, and the rule (at least 0.30 and the bound above zero) is **not met**. Without the four wars with a coalition of four or more states (the Gulf War, Kosovo, Afghanistan, Iraq) the relationship disappears: +0.152 (n = 14). For the wars of 1946–1973, where the era overlaps with CDB90 and there is no independence, Spearman is +0.679. This agrees with the literature: part of the effect is asymmetric wars of rich states against poor ones.

> **Conclusion.** This is not a refutation: 24 cases in 9 dependent dyads give too little power, while on the COW wars the direction matches but the relationship rests on wars with coalitions. Therefore country features (GDP per capita, democracy, mechanisation) are not entered into the index: they capture part of effectiveness in the 1900–1973 era and have not yet shown skill on modern wars. For the effectiveness of modern armies, expert assessments are still needed (Delphi panel, modules M5–M6).

# 5. Analysis of common claims

Language model answers and competitor materials about a “complete model” with hundreds of parameters contain errors that are useful to know about:

| Claim | What actually happens |
|---|---|
| “A complete mathematical model of the TENET MCI” with formulas for intelligence and psychology | the MCI core is closed (proprietary); such “syntheses” are language-model generalisations, not the company's formulas |
| “More units are more advantageous than new platforms” (reference to Atkinson, Kress, MacKay 2021) | in the work: more units and better weapons are more advantageous than improved targeting and the removal of deployment constraints |
| “A 10% advantage gives 21% power” | true only under the square law; in battles $\alpha \approx 0.65–1.0$ |
| “Intelligence acts as the square root of superiority” (Coulson 2019) | this is a consequence of the square law, not a law in its own right; in the data, intelligence ≈ a twofold advantage per step |
| “Artillery — linear law, tanks — quadratic law” | the linear law — fire without a visible target; artillery with drone adjustment conducts aimed fire |
| “Numbers decide more than is commonly believed” (TENET) | according to CDB90 the force ratio hardly predicts the outcome; leadership and morale weigh more |

# 6. Scientific novelty

## 6.1. Not new

- Lanchester-based military power index (Osipov 1915; TENET MCI; Dupuy QJM).
- Drones as a third force, cost exchange, depth of production (Gogerty 2026).
- Equations from the loss series of the war in Ukraine (Chung, Jeong 2024, SINDy on Oryx).
- Intelligence and morale in the equations (Deitchman, Shaffer, Shumov, Coulson, Atkinson–Kress –MacKay).

## 6.2. Possibly new (priority must be confirmed by a literature review)

1. **An index translated into force (M = exp(κ·MPI)), with out-of-sample validation according to pre-registered protocols**, including a negative result: open data predict neither the outcome nor the onset of wars. Publication of failures with pre-registration is almost unheard of in this field.
2. **α ≈ 1 only in the exploratory estimate:** the pre-registered α = 0.65 rejects both the linear and the square law; the apparent historical “law between linear and square” is explained by selection — attackers are qualitatively stronger. Check whether Hartley and Helmbold did this.
3. **The absolute intensity of combat does not follow any form of Lanchester** ($p \approx 0.16$, $q \approx 0.46$), while the loss ratio is consistent with $\alpha \approx 0.7–1$.
4. **Factor multipliers in units of numerical advantage** (leadership ≈ 9, morale ≈ 6, intelligence ≈ 2) and **the will threshold with censoring** (median 16%, morale × 1.29 per step).
5. **d for a specific war from its loss ratio and the model error σ calibrated on 10 wars.**
6. **Decomposition of the predictability of the outcome into potential and effectiveness out of sample** (almost all of it is effectiveness) and **the inverse order of an index of resources relative to the success of an offensive** across 10 wars (an exploratory result, $p \approx 0.03–0.07$).

## 6.3. Where the gap is

An attrition model with drones (output versus expenditure, cost exchange, EW), **validated out of sample**. The first check has been carried out: the simple relationship “stated output → loss ratio” according to Oryx 2022–2025 was not confirmed (section 2.3). The gap remains: inputs better than stated output are needed — expenditure of means, losses to EW, hit rate. Neither Gogerty nor Chung and Jeong performed out-of-sample validation.

# 7. Competitor: TENET Intelligence MCI

TENET Intelligence (United States) is an analytics company founded by former military and intelligence officers; its products are DAWN (defence market), MERIDIAN (national power and diplomatic influence) and SAGE (a single platform). Military Capability Index: “based on Lanchester's laws”, 150+ countries, 200+ indicators, four blocks — force structure, technological equipment, operational effectiveness, support capability. The equations are closed; there is no published validation against wars.

| Criterion | MPI Lanchester–Osipov | TENET MCI | Ahead |
|---|---|---|---|
| Country coverage | 8 (experts), 40 (pilot), 177 (extended set) | 150+ | TENET |
| Indicators | 253 (≈ 215 in the index) | 200+ | parity |
| Transparency | formulas, parameters, protocols and results are open; the core code is closed until version 1.0 | core closed | MPI |
| Combat law | α measured on battles, quadratic law rejected | claims $N^2$ | MPI |
| Historical testing | CDB90 and 10 wars, protocols fixed before the run | not published | MPI |
| Uncertainty | intervals, Monte Carlo, Sobol indices, model σ | not visible | MPI |
| Intelligence and will | multipliers and threshold measured on battles | declared only | MPI |
| Drones and production | out-of-sample validation: the relationship “output → losses” was not confirmed | articles, UAV market | open |
| Navy and air defence | salvo model (in progress) | unknown | MPI, after implementation |
| Nuclear weapons | separate axis | unknown | MPI |
| Coalitions | model with reliability, force share, delay, distance | not public | MPI |
| Flashpoint monitor | AUC 0.962 out of sample; the simple baseline (border, history, last year) gives 0.958 | MERIDIAN, untested | MPI, for honesty of validation |
| Forecast honesty | acknowledged: war outcomes are not predicted better than the uniform forecast | marketing without testing | MPI on trust |
| Data | open sources + provisional AI estimate (class D), without IISS | probably paid and closed OSINT | TENET |
| Expertise | Delphi panel not conducted | staff of former military & intelligence officers | TENET |
| Update and product | prototype, manual updating | SaaS platform | TENET |
| Market | none | defence industry clients, advisers | TENET |
| Cost | free data | paid | MPI |

**Bottom line.** The scientific part — testability, measured laws, honest uncertainty — is stronger in the MPI; data, coverage, expertise, product and market — in TENET. To close the gap: expert panel, 150+ countries, paid data when a budget appears, platform (FastAPI, React).

# 8. Plan

7. An attrition model with drones using better inputs than stated output (expenditure of means of destruction, cost exchange, degradation from EW): the simple relationship “output → loss ratio” has been tested and not confirmed (section 2.3); a new protocol before computation.
8. Salvo model for navies and air defence (depth of interceptor ammunition load, cost of interception).
9. The effectiveness multiplier E based on CDB90 factors goes into the “potential × effectiveness” branch of the theory; the sources of factor values for modern armies are the expert panel (modules M5, M6).
10. Literature review to confirm the priority of items 6.2 (Hartley, Helmbold, Lucas and Turkes, Dupuy).
11. Test effectiveness proxies on a base with many comparable clashes between states (for example, ACLED) — a separate protocol before the calculation.

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